

Lab 3 - Architecture Selection & Written Justification

Inter-City Freight Load Matching Marketplace

Architecture Selection

We chose Microservices Architecture for the Inter-City Freight Load Matching Marketplace.

Architectural Choice

Microservices is suitable because the marketplace contains distinct functions including freight-load management, competitive bidding, carrier verification, shipment tracking, electronic proof-of-delivery (e-POD) verification, escrow payment processing, data storage, and notifications. These functions can be separated into independently managed services that communicate through well-defined interfaces.

Reason 1 - Independent Scaling and Performance

The bidding and carrier-verification service may experience heavy activity when many carriers compete for posted loads. It can be scaled independently from notification or shipment-tracking services. This is particularly useful because the system requires escrow milestone payments to be disbursed within 60 seconds after verified e-POD submission.

Reason 2 - Fault Isolation and Modularity

A failure in a secondary service, such as the notification service, should not prevent load posting, bidding, or shipment processing. Separating services also makes it easier to modify individual functions such as carrier verification, tracking, or payment processing without redesigning the entire marketplace.

Security Advantage

Each service can enforce authentication and authorization at its API boundary. Sensitive transaction and payment operations can be isolated from general user-facing services, while communication between services can use secure channels. This supports transaction integrity for bids, carrier selection, milestone updates, e-POD verification, and escrow payment release.

Performance Benefit

Services can process requests independently, reducing unnecessary work in user-facing components. High-load functions such as bidding, verification, and payment processing can be scaled separately, improving responsiveness and helping the system satisfy its payment-disbursement performance target.